EXERCISE – 10

I. Choose the best answer:

1. The mean and variance of a Bernoulli distribution with usual notations are

(a) $p, \sqrt{pq}$ (b) p, pq (c) np, npq (d) λ

2. In getting a `6' in a throw of a die the Bernoulli distribution

(a) $\left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{p-x}$ (b) $\left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{1-x}$ (c) $\left(\frac{5}{6}\right)^x \left(\frac{1}{6}\right)^{1-x}$ (d) none

3. The mean and variance of $B(n, p)$ are

(a) npq, np (b) $np, \sqrt{npq}$ (c) np, npq (d) $\sqrt{npq}, np$

4. In B (n, p) the probability `n' successes is

(a) $nC_x p^x q^{n-x}$ (b) 1 (c) p^n (d) q^n

5. If $X \sim B(n_1, p)$ is independent of $Y \sim B(n_2, p)$ then $X + Y$ is distributed as

(a) B $(n_1 + n_2, 2p)$ (b) B (n_1, p) (c) B $(n_1 + n_2, p)$ (d) B(n_2, p)

6. If the expectation of a Poisson variable is `1' then $P(x < 1)$ is

(a) e^{-1} (b) $1-2e^{-1}$ (c) $1-\frac{5}{2e^{-1}}$ (d) none

7. Poisson distribution is a limiting case of Binomial distribution when

(a) $n \to \infty$; $p \to 0$ and $np = \sqrt{m}$ (b) $n \to 0$; $p \to \infty$ and $p = 1/m$

(c) $n \to \infty$; $p \to \infty$ and $np = m$ (d) $n \to \infty$; $p \to 0$ and $np = m$

8. Poisson distribution corresponds to

(a) Rare events (b) Certain event

(c) Impossible event (d) Almost sure event

9. If $\left(\frac{2}{3} + \frac{1}{3}\right)^9$ represents a Binomial distribution its standard deviation is

(a) 2 (b) $\sqrt{2}$ (c) 6 (d) 3

10. In a Poisson distribution

(a) Mean = Variance (b) Mean < Variance

(c) Mean > Variance (d) Mean ≠ Variance

BDHK8X

11. If X ~ U (-5, 5) then its p.d.f. f(x) in $-5 < x < 5$ is

(a) $\frac{1}{5}$ (b) $\frac{1}{2 \times 5}$ (c) $\frac{1}{5^2}$ (d) $\frac{1}{2}$